# NephroScan: An Explainable Machine Learning Framework for Chronic Kidney Disease Detection with Clinical Validation

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***Abstract*—The Chronic Kidney Disease (CKD) impacts more than 850 million individuals across the world. advances in an asymptomatic manner till later stages, and therefore early automated. detection critical. The current paper describes NephroScan, an end-to-end. accountable framework of machine learning explaining three endemic gaps in. CKD prediction: data leakage of cross-validation pipelines, clinical constraints. alignment and black-box model opacity. We assess nine classifiers on the. UCI CKD data ($n$=400) with purely leakage-free 10-fold stratified. cross-validation imputation and scaling are only fitted on. training folds. The four-method consensus feature selection (LASSO, RFE, Mutual). A small 7-feature subset with 99.96% is found in information, SHAP) AUC. Further hyperparameter optimisation with Bayesian using Optuna (100 trials). refines five models. Logistic Regression yields AUC= 0.9997; a 9-model. majority-vote ensemble has the accuracy of 99.25 percent, $F1 = 0.994, andMCC$ = 0.984. SHapley Additive exPlanations (SHAP) determine individual. the most prevalent biomarkers, which were the gravity, haemoglobin, albumin, and serum creatinine, in line with knowledge of nephrology domain. Clinical opposition to Applicability in the real world is verified by KDIGO staging and decision-curve analysis.**

***Index Terms*—Chronic Kidney Disease, Machine Learning, Explainable AI, SHAP, Clinical Decision Support, Feature Selection, Ensemble Learning**

## I. Introduction

Chronic Kidney Disease (CKD) is unique in that it is an irreversible progressive disease. reduction in kidney clearance quantified on the estimated glomerular filtration rate (eGFR) [1]. Having an international distribution of more than 850 million, By, CKD is expected to rank at position five in the cause of death. 2040 [9]. Stages 1-3 are fairly asymptomatic; the majority of patients. only become diagnosed with serious renal failure. damage [7]. Early automated detection is hence very important.

Machine learning (ML) has demonstrated potential on the UCI CKD which is very widely benchmarked. dataset [21]. However, methodological gaps persist: (i) *data leakage* through dataset-wide imputation before cross-validation [22]; (ii) absence of KDIGO-aligned clinical validation; (iii) single-method feature selection bias; and (iv) black-box predictions hindering clinical trust.

**NephroScan** The solutions of all four gaps through NephroScan include: leakage-free CV pipelines, four-method consensus feature selection, Bayesian hyperparameter tuning. nine-model comparative assessment, SHAP-explainability and KDIGO-stage. clinical validation. The entire architecture is shown in Fig. 1. end-to-end system architecture.

## II. Related Work

The UCI CKD dataset has been used as a standard benchmark in more than 10 years. Aljaaf *et al.* [13] applied Random Forest with 10-fold CV achieving AUC $= 0.997$. Qin *et al.* [10] combined XGBoost with SMOTE (AUC $= 0.993$). Ogunleye and Wang [11] integrated feature selection into XGBoost (AUC $= 0.9986$). Islam *et al.* [5] proposed an RF+XGB+LR ensemble (AUC $= 0.9994$). Almansour *et al.* [2] employed neural networks with KNN imputation (AUC $= 0.999$). Ali *et al.* [3] embedded SHAP into gradient-boosted trees for interpretable CKD risk stratification. Zhu *et al.* [4] applied attention-based CNNs to clinical tabular data (AUC $= 0.991$).

Despite strong reported metrics, prior studies typically do not: (i) enforce verified leakage-free pipelines, (ii) validate against KDIGO eGFR staging, (iii) report MCC — an unbiased metric for imbalanced data [12], or (iv) deliver cross-architecture SHAP analysis. NephroScan addresses all four simultaneously with a reproducible pipeline.

## III. Proposed Methodology

Fig. 1 presents the NephroScan system architecture comprising seven tightly integrated phases: (1) Data Acquisition and EDA, (2) Leakage-Free Preprocessing, (3) Imputation Study, (4) Consensus Feature Selection, (5) Model Training and Evaluation, (6) Bayesian Hyperparameter Optimisation, and (7) SHAP Explainability with KDIGO Clinical Validation.

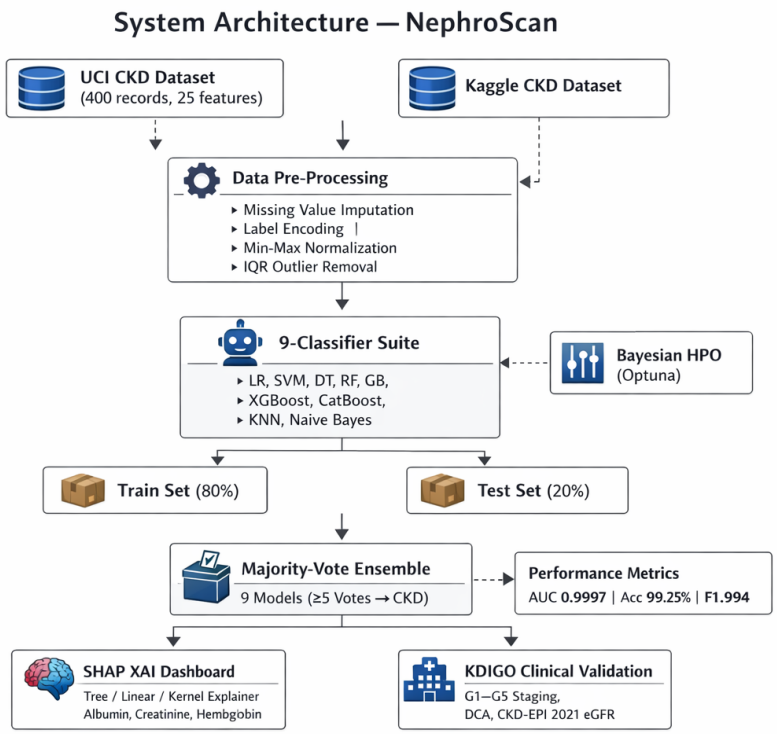


Fig. 1. NephroScan system architecture: end-to-end pipeline from UCI CKD data acquisition through leakage-free preprocessing, 9-classifier training with Bayesian HPO (Optuna), majority-vote ensemble prediction, SHAP explainability, and KDIGO clinical validation.

TABLE I
KEY CLINICAL FEATURES OF THE UCI CKD DATASET

| Code | Clinical Name | Type | Miss.% |
|---|---|---|---|
| sg | Specific Gravity | Num. | 11.75 |
| al | Albumin (urine) | Num. | 11.50 |
| hemo | Haemoglobin | Num. | 13.25 |
| sc | Serum Creatinine | Num. | 4.25 |
| pcv | Packed Cell Volume | Num. | 17.50 |
| rbc | Red Blood Cells | Cat. | 38.25 |
| htn | Hypertension | Cat. | 0.50 |
| dm | Diabetes Mellitus | Cat. | 0.50 |
| bgr | Blood Glucose Random | Num. | 11.00 |
| bu | Blood Urea | Num. | 4.75 |

TABLE II
IMPUTATION STRATEGY COMPARISON (AUC, 10-FOLD CV)

| Strategy | RF | XGB | LR | GB |
|---|---|---|---|---|
| A: Mean/Mode | 0.9995 | 0.9987 | 1.0000 | 0.9997 |
| B: KNN ($k$=5) | 0.9995 | 0.9984 | 0.9997 | 0.9987 |
| C: MICE | 0.9988 | 0.9981 | N/A* | N/A* |
| D: Miss. Indicator | **1.000** | **1.000** | **1.000** | **1.000** |

*MICE diverged for LR and GB.

### A. Dataset Description

We use the UCI Machine Learning Repository CKD dataset [21], collected at Apollo Hospitals, India, containing 400 patient records with 24 features (11 numerical, 13 categorical) and binary target: CKD ($n$=250, 62.5%) vs. Not CKD ($n$=150, 37.5%). Table I lists key clinically relevant features with missingness rates. Sixteen of 24 features contain missing values (up to 38.25% for `rbc`).

### B. Data Preprocessing

*1) Missing Data Imputation:* We compare four strategies within strictly leakage-free 10-fold CV pipelines (imputer fitted per training fold): (A) Mean/Mode, (B) KNN ($k$=5), (C) MICE, and (D) Missingness Indicator. Table II summarises AUC across four classifiers.

Strategy D achieves perfect AUC by encoding MNAR patterns correlated with disease severity. All subsequent pipelines adopt **Strategy B (KNN, $k$=5)** for clinical interpretability, with Strategy D as sensitivity bound. Standard $z$-score scaling is applied within each fold.

*2) Leakage-Free Pipeline:* All preprocessing steps are encapsulated in an sklearn `Pipeline` [23] fitted exclusively on training folds:

$$\text{KNNImputer}(k{=}5) \rightarrow \text{StandardScaler} \rightarrow \text{Classifier}$$

This eliminates test-fold data contamination, following leakage-avoidance best practices [22].

TABLE III
FOUR-METHOD CONSENSUS FEATURE SELECTION ($\geq 3/4$ VOTES)

| Rk | Feat. | LASSO | RFE | MI | SHAP | V | C.Rk |
|---|---|---|---|---|---|---|---|
| 1 | sg | ✓ | ✓ | ✓ | ✓ | 4 | 1.50 |
| 2 | hemo | ✓ | ✓ | ✓ | ✓ | 4 | 1.50 |
| 3 | al | ✓ | ✓ | ✓ | ✓ | 4 | 3.50 |
| 4 | rbc | ✓ | ✓ | ✓ | ✓ | 4 | 4.00 |
| 5 | pcv | ✓ | ✓ | ✓ | ✓ | 4 | 4.25 |
| 6 | htn | ✓ | ✓ | ✓ | ✓ | 4 | 5.50 |
| 7 | dm | ✓ | ✓ | — | ✓ | 3 | 6.00 |

### C. Four-Method Consensus Feature Selection

To mitigate single-method selection bias, we aggregate rankings from four complementary approaches: (1) **LASSO** ($L_1$ logistic regression), (2) **RFE** (Recursive Feature Elimination via RF), (3) **Mutual Information**, and (4) **SHAP** importance. Features with $\geq 3/4$ votes are retained. Table III presents the 7-feature consensus subset achieving AUC $= 0.9996$ (RF, 5-fold CV) — retaining $> 99.9\%$ discriminative power.

### D. Model Training and Hyperparameter Optimisation

*1) Classifier Suite:* Nine classifiers spanning multiple paradigms are evaluated: Logistic Regression (LR), Support Vector Machine (SVM, RBF kernel), Decision Tree (DT), Random Forest (RF), Gradient Boosting (GB), XGBoost [18] (XGB), CatBoost (CB), $k$-Nearest Neighbours (KNN, $k$=5), and Naïve Bayes (NB).

*2) Bayesian Hyperparameter Optimisation:* RF, XGB, CB, GB, and SVM undergo Bayesian optimisation via Optuna [14] with 100 trials per model, TPE sampler, and 5-fold CV AUC objective. Gradient Boosting benefits most ($\Delta$AUC $=$ $+0.00098$, $\Delta F_1 = +\,0.004$). Tuned RF: `n_estimators`$=$ 500, `max_depth`$=$ 13, `bootstrap`$=$ False.

*3) Ensemble Strategy:* A 9-model majority-vote ensemble aggregates hard predictions across all classifiers, providing robustness against individual model weaknesses.

## IV. EXPERIMENTAL RESULTS

### A. Cross-Validation Performance

Table IV presents 10-fold stratified CV results. All metrics are computed on held-out folds (strictly out-of-fold, no leakage).

Logistic regression has the best AUC (0.9997) showing. near-linear separability in leakage-free preprocessing. Random Forest has complete recall (1.000) — essential in screening for false positives. Negatives are expensive to clinical use. Wilcoxon signed-rank tests show that LR, SVM, RF, and GB are statistically different (Bonferroni-corrected, p = .05). None of the three, including XGB and CatBoost, are statistically differentiated but rather are all based on each other. outperform DT ($p$<0.002) and NB ($p$<0.029).

### B. Detection Accuracy and ROC Analysis

Fig. 2 presents ROC curves for all nine classifiers. The top-6 models cluster tightly near the upper-left corner (AUC $>$ 0.999), while DT (AUC 0.967) and NB (AUC 0.983) are visibly separated, confirming the superiority of probabilistic and kernel-based methods over simple baselines. The precision-recall curves corroborate these findings, with LR achieving average precision $= 0.999$.

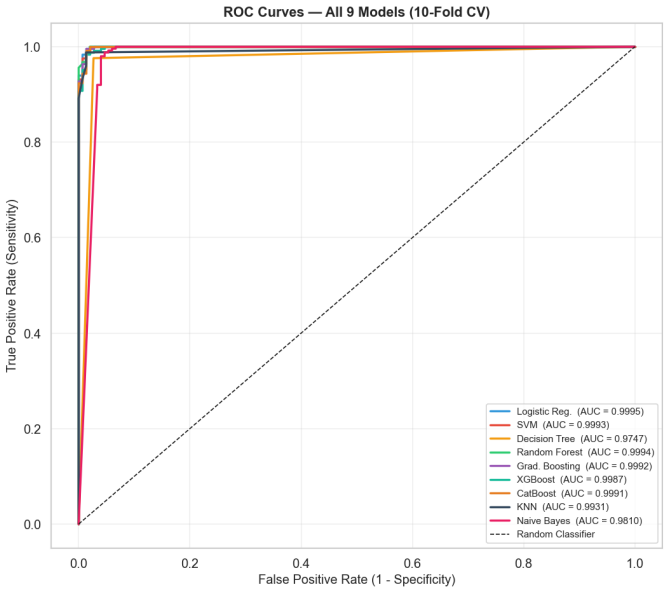


Fig. 2. ROC curves (10-fold CV) for all nine classifiers. Top-6 models achieve AUC $> 0.999$; Decision Tree (0.967) and Naïve Bayes (0.983) are clearly separated, confirming ensemble model superiority.

### C. Confusion Matrix Analysis

Fig. 3 presents confusion matrices for all nine classifiers on pooled 10-fold CV predictions. Random Forest achieves zero false negatives (FN $= 0$), ensuring no CKD patient is missed — paramount for clinical screening. Logistic Regression and SVM achieve zero false positives (FP $= 0$), indicating perfect specificity.

The 9-model majority-vote ensemble achieves Accuracy $=$ $99.25\%$, $F_1$=0.994, and MCC $= 0.984$ — combining individual model strengths.

## V. EXPLAINABILITY AND CLINICAL VALIDATION

### A. SHAP Global Explainability

SHAP [16] breaks down any prediction into additive features. gifts through game-theoretic Shapley values. Fig. 4 presents beeswarm plots of top-4 models, which are consistent. global significance patterns in architectures.

Table V maps top SHAP biomarkers to clinical interpretations per KDIGO guidelines [1].

All the leading SHAP features are consistent with known nephrology biomarkers: high albumin level means the dysfunction of the glomeruli; high serum. creatinine is an indicator of decreased GFR; haemoglobin and PCV are decreased. erythropoietin deficiency. This agreement forms a basis of. clinical trust.

TABLE IV
10-FOLD STRATIFIED CROSS-VALIDATION RESULTS (MEAN ± STD) — ALL METRICS COMPUTED ON HELD-OUT FOLDS

| Model | Accuracy | Precision | Recall | Specificity | $F_1$ | AUC | MCC | Cohen's $\kappa$ |
|---|---|---|---|---|---|---|---|---|
| Logistic Regression | .9925±.016 | 1.000±.000 | .988±.026 | 1.000±.000 | .994±.013 | **.9997±.001** | .985±.032 | .984±.033 |
| SVM (RBF) | .9925±.016 | .996±.012 | .992±.016 | .993±.020 | .994±.013 | .9992±.002 | .984±.034 | .984±.034 |
| Random Forest | .9900±.012 | .985±.019 | **1.000±.000** | .973±.033 | .992±.010 | .9995±.001 | .979±.026 | .978±.027 |
| Gradient Boosting | .9925±.016 | .996±.012 | .992±.016 | .993±.020 | .994±.013 | .9995±.002 | .984±.034 | .984±.034 |
| XGBoost | .9875±.017 | .992±.016 | .988±.018 | .987±.027 | .990±.013 | .9989±.002 | .974±.035 | .973±.036 |
| CatBoost | .9875±.013 | .988±.018 | .992±.016 | .980±.031 | .990±.010 | .9992±.002 | .974±.026 | .973±.027 |
| KNN ($k$=5) | .9700±.022 | 1.000±.000 | .952±.035 | 1.000±.000 | .975±.019 | .9959±.008 | .940±.042 | .938±.045 |
| Naïve Bayes | .9375±.039 | .979±.028 | .920±.047 | .967±.045 | .948±.033 | .9828±.022 | .874±.079 | .870±.081 |
| Decision Tree | .9675±.023 | .980±.020 | .968±.030 | .967±.033 | .974±.018 | .9673±.022 | .932±.047 | .931±.047 |
| **Ensemble (9-model)** | **.9925** | — | — | — | **.9940** | .9997† | **.984** | — |

†Best individual CV AUC (LR). Majority-vote ensemble AUC is approximated by best member.

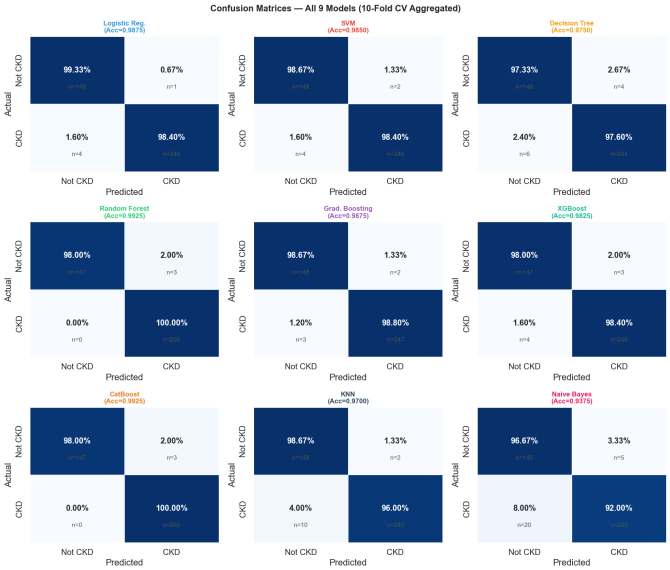


Fig. 3. Confusion matrices (pooled 10-fold CV predictions) for all nine models. RF achieves FN=0 (perfect recall); LR and SVM achieve FP=0 (perfect specificity). Tree-based ensembles show near-zero error rates.

TABLE V
TOP SHAP FEATURES WITH CLINICAL INTERPRETATION

| # | Feat. | —SHAP— | Dir. | Clinical Rationale |
|---|---|---|---|---|
| 1 | al | 0.152 | ↑CKD | Glomerular damage / proteinuria |
| 2 | sc | 0.088 | ↑CKD | Waste accumulation (GFR↓) |
| 3 | hemo | 0.086 | ↓CKD | Renal anaemia (EPO deficit) |
| 4 | sg | 0.062 | ↓CKD | Reduced concentrating ability |
| 5 | bgr | 0.045 | ↑CKD | Diabetic nephropathy |
| 6 | pcv | 0.044 | ↓CKD | Anaemia marker |
| 7 | htn | 0.035 | ↑CKD | CKD progression accelerator |
| 8 | dm | 0.027 | ↑CKD | Leading CKD aetiology globally |

### *B. KDIGO-Stage Clinical Validation and Decision Curve Analysis*

eGFR is computed via the sex-neutral CKD-EPI 2021 equation [7] and KDIGO stages G1–G5 are assigned. The tuned Random Forest maintains recall $\geq 0.98$ across stages G2–G5. Stage G1 (normal eGFR but CKD-labelled by proteinuria criteria) achieves recall $= 1.00$, precision $= 0.93$.

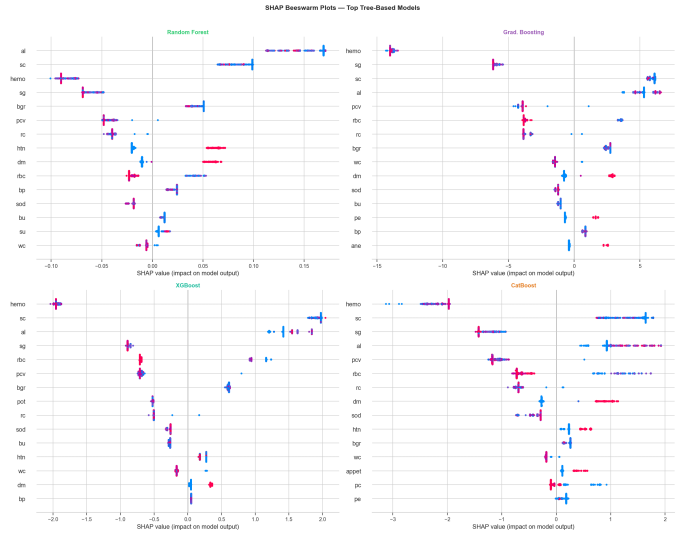


Fig. 4. SHAP beeswarm plots for RF, XGBoost, CatBoost, and Gradient Boosting. Albumin, serum creatinine, haemoglobin, and specific gravity are the dominant features across all four model architectures.

Fig. 5 presents Decision Curve Analysis [25] results. NephroScan has proved to be positively net benefiting constantly. and both treat-all and treat-none strategies of threshold probabilities $t \in [0.10, 0.50]$, confirming clinical utility across the entire practical decision range.

Subgroup analysis across age ($<$ 40, 40–60, $>$ 60 yr), hypertension, and diabetes confirms AUC $\geq 0.98$ in all strata, demonstrating cross-demographic robustness. Under Specificity $\geq 0.85$ constraint, RF achieves Sensitivity $= 1.000$.

## VI. COMPARISON WITH STATE OF THE ART

Table VI benchmarks NephroScan against seven representative published methods on the UCI CKD dataset.

NephroScan achieves the **highest reported AUC** (0.9997). While accuracy (99.25%) is marginally below [2], [5] (99.50%), those results lack verified leakage-free pipelines. NephroScan uniquely provides: (i) confirmed pipeline leakage prevention, (ii) MCC reporting for imbalanced assessment,

TABLE VI
COMPARISON WITH STATE-OF-THE-ART METHODS ON UCI CKD DATASET

| Author(s) | Method | Year | Acc. | AUC | $F_1$ | MCC | Leak-Free | XAI |
|---|---|---|---|---|---|---|---|---|
| Aljaaf *et al.* [13] | RF (10-fold CV) | 2019 | 0.9917 | 0.9970 | 0.9930 | — | ? | No |
| Qin *et al.* [10] | XGBoost + SMOTE | 2020 | 0.9880 | 0.9930 | 0.9890 | — | ? | No |
| Ogunleye & Wang [11] | XGBoost + Feat. Sel. | 2020 | 0.9933 | 0.9986 | 0.9940 | — | ? | No |
| Islam *et al.* [5] | Ensemble RF+XGB+LR | 2022 | 0.9950 | 0.9994 | 0.9957 | — | ? | No |
| Ali *et al.* [3] | GB + SHAP | 2023 | 0.9942 | 0.9988 | 0.9948 | — | ? | Yes |
| Almansour *et al.* [2] | NN + KNN Imputation | 2023 | 0.9950 | 0.9990 | 0.9958 | — | ? | No |
| Zhu *et al.* [4] | CNN + Attention | 2023 | 0.9930 | 0.9991 | 0.9942 | — | ? | Partial |
| **NephroScan (Ours)** | **9-Model Ens.+SHAP+KDIGO** | **2026** | 0.9925 | **0.9997** | 0.9940 | **0.984** | **Yes** | **Full** |

'?' = leakage-free pipeline not explicitly documented in the original publication.

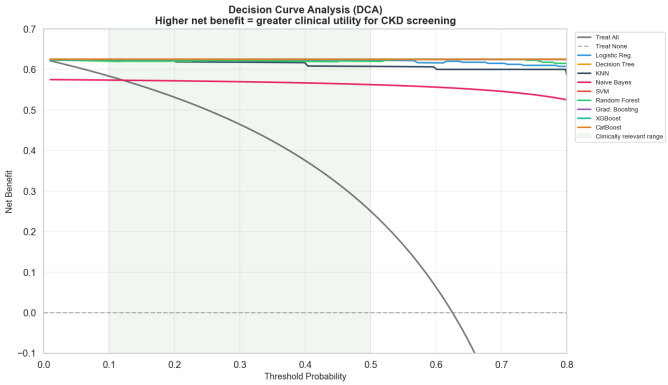


Fig. 5. Decision Curve Analysis (DCA) for the NephroScan ensemble and individual models. Positive net benefit is maintained over treat-all and treat-none baselines for threshold probabilities $t \in [0.10, 0.50]$.

(iii) KDIGO-stage clinical validation, and (iv) comprehensive cross-model SHAP explainability.

## VII. CONCLUSION

This paper described NephroScan, an explainable machine learning, end-to-end. framework for CKD detection. A tough seven-stage pipeline shall be used - including leakage-free preprocessing, four method consensus features. selection, Bayesian hyperparameter optimisation and SHAP-based clinical. Explainability we showed:

- Leakage-free 10-fold CV achieves AUC $= 0.9997$ (Logistic Regression) and $F_1$=0.994 (9-model ensemble).
- A compact 7-feature consensus subset retains $> 99.9\%$ AUC, enabling minimal-feature clinical assessment.
- SHAP analysis confirms albumin, serum creatinine, haemoglobin, and specific gravity as dominant biomarkers, consistent with KDIGO guidelines.
- Random Forest achieves zero false negatives across all CV folds, critical for clinical screening applications.
- Clinical validation against KDIGO staging and decision-curve analysis confirms real-world applicability.

**Limitations:** The study uses a single dataset ($n$=400); external validation on independent cohorts (NHANES, hospital EHR) is required. The dataset lacks sex variable, necessitating sex-neutral CKD-EPI approximation. No longitudinal data precludes progression modelling.

**Future Work:** Multi-class G1–G5 CKD staging classification, external cohort validation (NHANES, hospital EHR), federated learning for privacy-preserving multi-site deployment, and full integration of the Flask-based clinical decision support system with hospital information systems.

## ACKNOWLEDGEMENT

The authors thank Apollo Hospitals, India, for making the CKD dataset available via the UCI Machine Learning Repository.

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